EXPT NO:3	CLEANSE DATA BY HANDLING MISSING VALUES, OUTLIERS, AND INCONSISTENCIES USING TABLEAU AND POWERBI
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What are the potential impacts of missing values in a dataset, and what are some common strategies to address them in Tableau and Power BI?

   Missing values can reduce the accuracy of analysis, produce misleading visualizations, and affect the reliability of reports and predictions. In Tableau and Power BI, missing values can be handled by filtering them out, replacing them with default values using calculated fields, filling null values in Power Query, or using statistical measures such as the mean or median where appropriate. Proper handling of missing values improves data quality and analysis.

2. What is the importance of identifying and addressing outliers in a dataset? How can outliers affect data analysis and model performance?

   Outliers are unusually high or low values that differ significantly from the rest of the data. They can distort averages, trends, and visualizations, leading to inaccurate conclusions and reduced prediction accuracy. Identifying and treating outliers helps improve the reliability of data analysis and enhances the performance of analytical and predictive models.

3. What types of data inconsistencies might you encounter during data cleansing, and how can Tableau and Power BI help in resolving them?

   Common data inconsistencies include duplicate records, missing values, incorrect data types, inconsistent text formats, spelling errors, and invalid date formats. Tableau and Power BI provide data preparation features such as filtering, data type conversion, removing duplicates, replacing values, and Power Query transformations to clean and standardize the data before analysis.

4. How do you identify patterns in data using visualizations to spot missing values, outliers, and inconsistencies?

   Visualizations such as scatter plots, box plots, bar charts, line charts, and heat maps help identify unusual patterns in data. Missing values may appear as gaps, outliers as isolated points far from other observations, and inconsistencies as unexpected variations or irregular trends. These visualizations make it easier to detect data quality issues before performing analysis.

5. What is the role of calculated fields in data cleansing, and how can they be used to handle missing values and outliers in Tableau/Power BI?

   Calculated fields allow users to create new values or modify existing data without changing the original dataset. They can be used to replace null values, calculate averages, identify outliers using conditional logic, and create custom metrics such as prediction error. This helps improve data quality and ensures more accurate visualizations and reports

IN-LAB EXERCISE:

OBJECTIVE:

To clean a given dataset (e.g., Superstore Sample in Tableau and Financials/Sales Data in Power BI) by identifying and resolving missing values, outliers, and inconsistencies using built-in tools, ensuring the dataset is accurate and ready for analysis.

PROCEDURE:

1. **Open the Tool and Connect Dataset**
   - Open Tableau Desktop , Power BI Desktop.
   - Load the built-in dataset (Sample - Superstore in Tableau / Financials in Power BI).
   - Preview the dataset to understand its structure.
2. **Initial Inspection**
   - Explore fields such as Sales, Profit, Region, or Product Category.
   - Check for nulls, inconsistent text values (e.g., "South Region" vs. "South"), or date formatting issues.
3. **Handling Missing Values**
   - Use Data Interpreter (Tableau) / Power Query (Power BI) to highlight nulls.
   - Replace missing numerical values with mean/median, and text with "Unknown".
   - Use calculated fields: IFNULL([Field], Default) in Tableau or IF(ISBLANK([Field]), "Unknown", [Field]) in Power BI.
4. **Detecting and Managing Outliers**
   - Create scatter or box plots for Sales and Profit to identify extreme values.
   - Choose strategies:
     - Filter them temporarily.
     - Apply caps using calculated columns.

        - Flag them with labels for review.

5. **Resolving Inconsistencies**
    - Use Group/Alias (Tableau) / Replace Values (Power BI) to standardize text.
    - Normalize data formats, such as date fields or currency units.
6. **Validation with Visuals**
    - Build dashboards with histograms, bar charts, and box plots to visually confirm improvements.
    - Compare before/after cleaning stages for accuracy.
7. **Save the Cleaned Dataset**
    - Save as .twbx (Tableau) / .pbix (Power BI).
    - Optionally export cleaned data to .csv.
8. **Summary Dashboard**
    - Create a visual summary of the cleaning process, highlighting:
        - Fields corrected.
        - Nulls/outliers removed or adjusted.
        - Clean dataset metrics (record count, completeness %).

Experiment Dataset:

1. Tableau: *Sample - Superstore*
2. Power BI: *Financials* or *Sales Data*

How to Access:

Tableau:

- **Navigate to:** *Start Page → Sample Workbooks*
- Or: *Help → Sample Data → Open Sample Data Folder*

Power BI:

- **Navigate to:** *Home → Get Data → Excel → Choose sample datasets (e.g., Financial Sample.xlsx)*
- Microsoft also hosts Power BI sample datasets online.

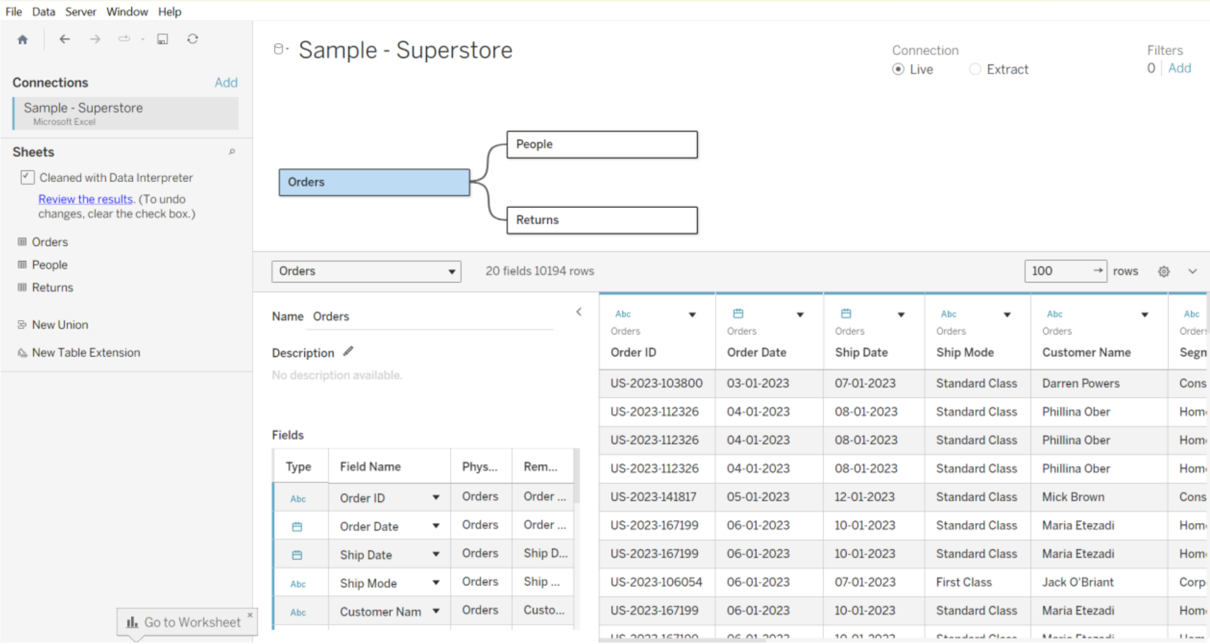
Fields Used:

1. Sales
2. Profit
3. Region
4. Category

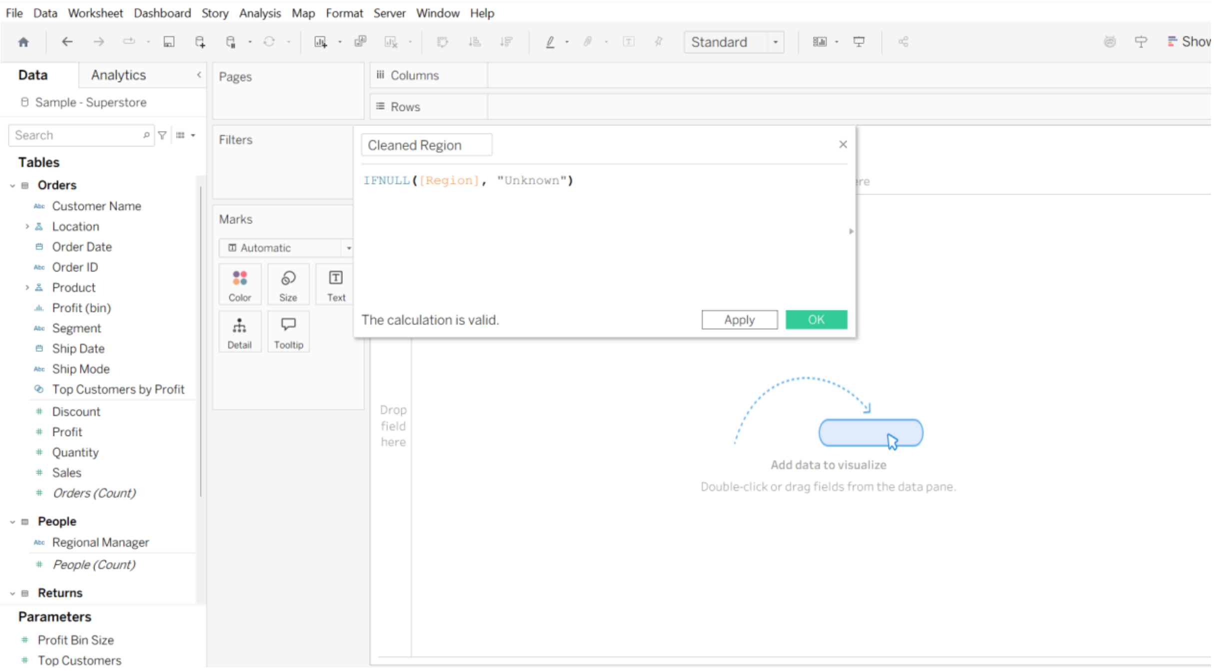
5. Order Date / Transaction Date

TABLEAU:

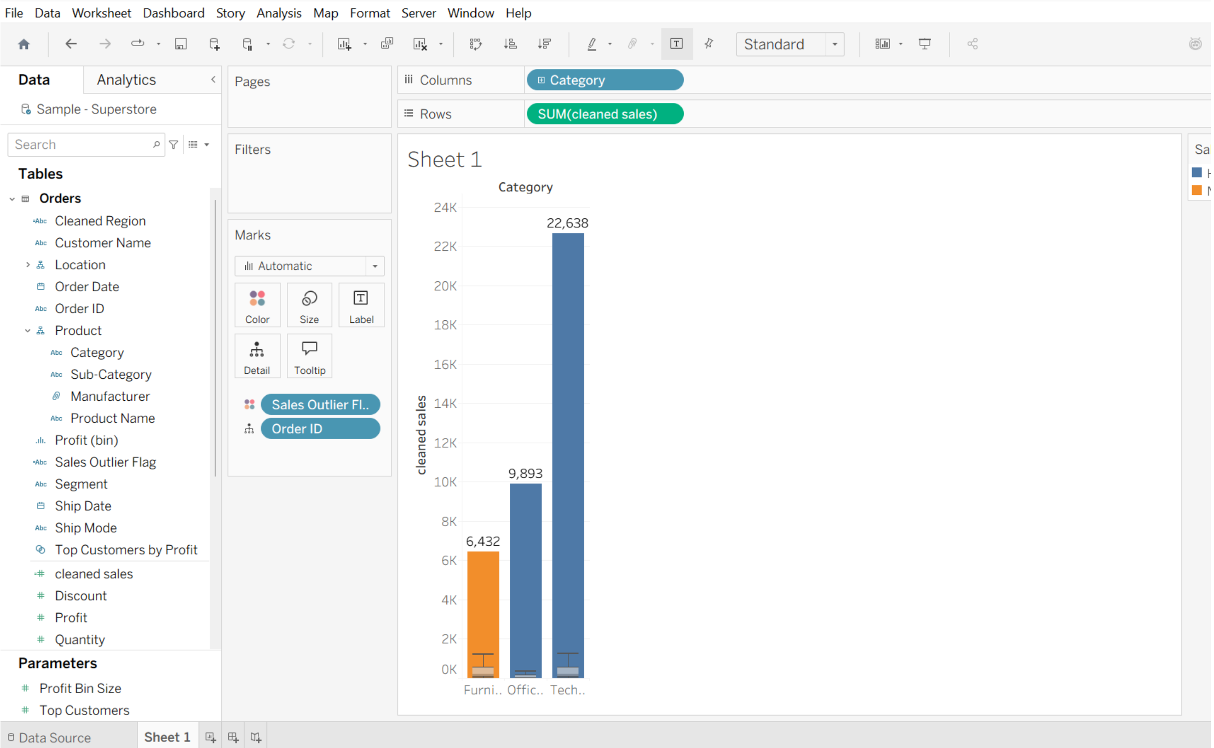
Connect & Inspection

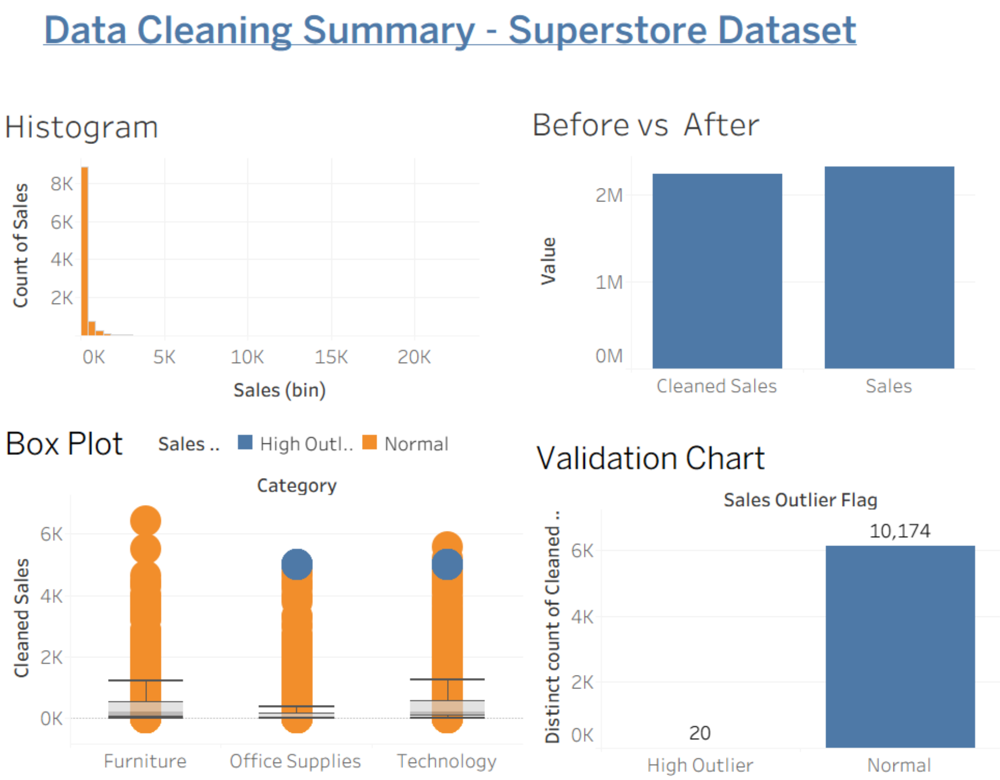


Handling Missing Values



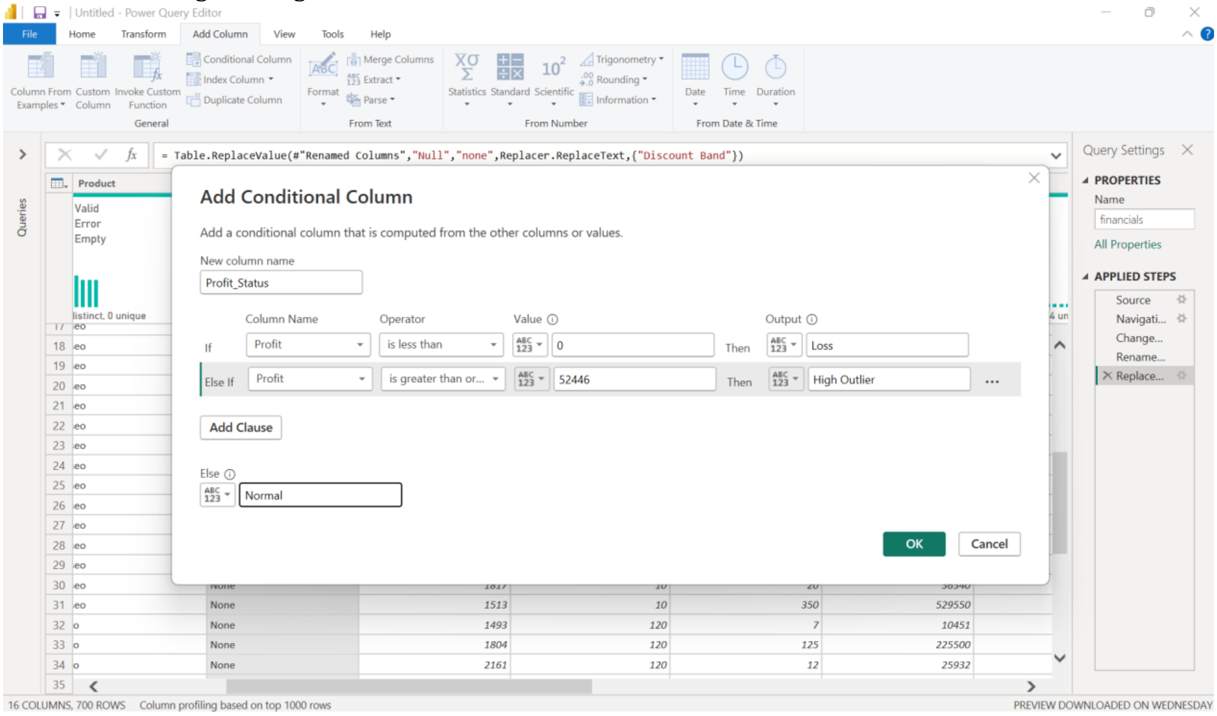
Detecting & Managing Outliers





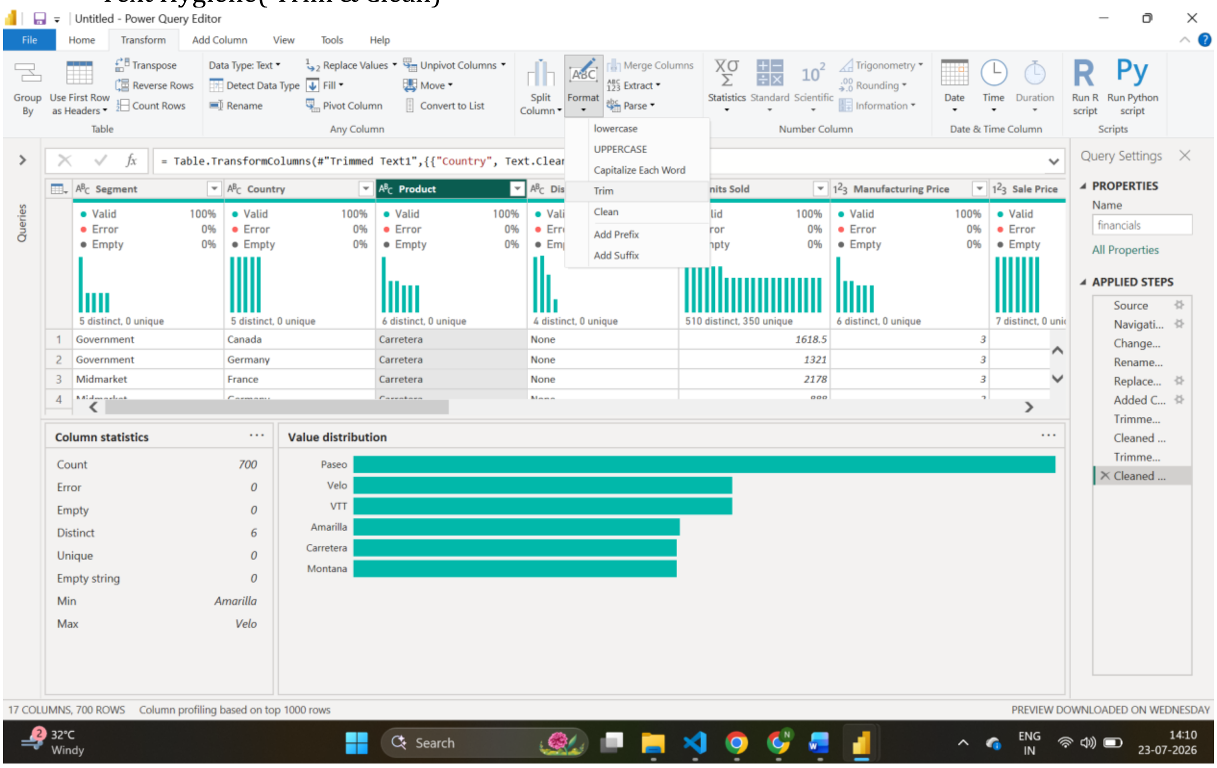
POWERBI:

- Handling Missing Values & Column Names

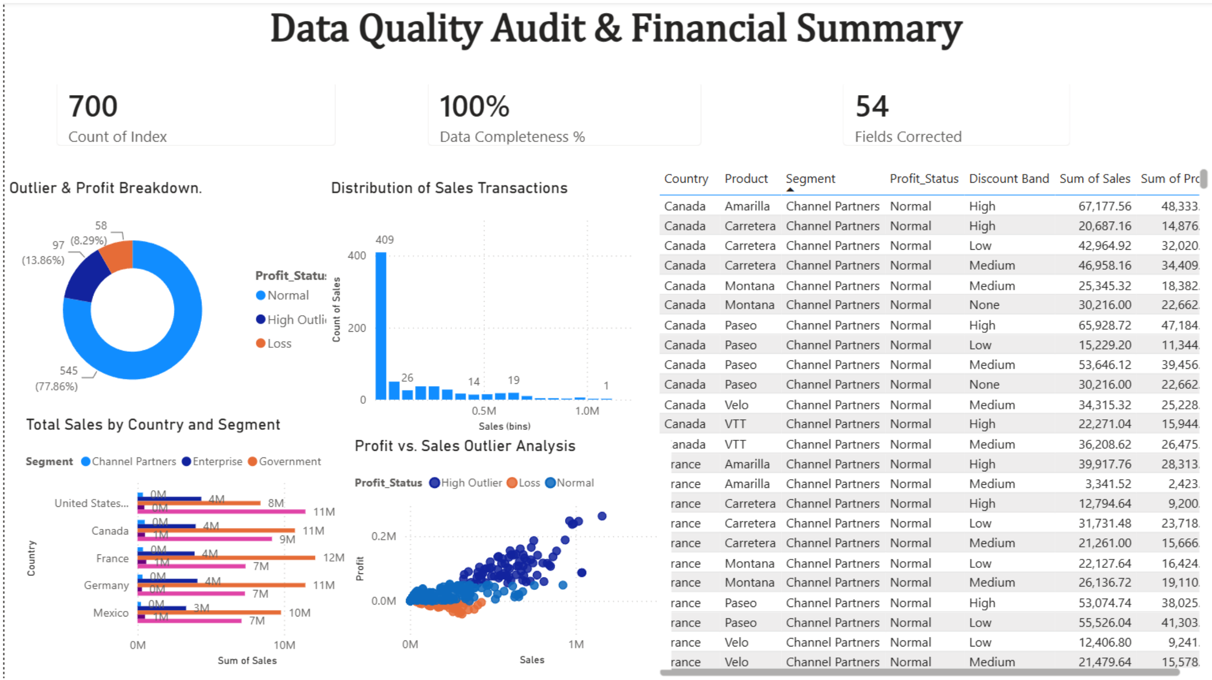


Resolving Inconsistencies & Formatting

- Text Hygiene(Trim & Clean)



DASHBOARD:



Country	Product	Segment	Profit_Status	Discount Band	Sum of Sales	Sum of Pro
Canada	Amarilla	Channel Partners	Normal	High	67,177.56	48,333.
Canada	Carretera	Channel Partners	Normal	High	20,687.16	14,876.
Canada	Carretera	Channel Partners	Normal	Low	42,964.92	32,020.
Canada	Carretera	Channel Partners	Normal	Medium	46,958.16	34,409.
Canada	Montana	Channel Partners	Normal	Medium	25,345.32	18,382.
Canada	Montana	Channel Partners	Normal	None	30,216.00	22,662.
Canada	Paseo	Channel Partners	Normal	High	65,928.72	47,184.
Canada	Paseo	Channel Partners	Normal	Low	15,229.20	11,344.
Canada	Paseo	Channel Partners	Normal	Medium	53,646.12	39,456.
Canada	Paseo	Channel Partners	Normal	None	30,216.00	22,662.
Canada	Velo	Channel Partners	Normal	Medium	34,315.32	25,228.
Canada	VTT	Channel Partners	Normal	High	22,271.04	15,944.
anada	VTT	Channel Partners	Normal	Medium	36,208.62	26,475.
rance	Amarilla	Channel Partners	Normal	High	39,917.76	28,313.
rance	Amarilla	Channel Partners	Normal	Medium	3,341.52	2,423.
rance	Carretera	Channel Partners	Normal	High	12,794.64	9,200.
rance	Carretera	Channel Partners	Normal	Low	31,731.48	23,718.
rance	Carretera	Channel Partners	Normal	Medium	21,261.00	15,666.
rance	Montana	Channel Partners	Normal	Low	22,127.64	16,424.
rance	Montana	Channel Partners	Normal	Medium	26,136.72	19,110.
rance	Paseo	Channel Partners	Normal	High	53,074.74	38,025.
rance	Paseo	Channel Partners	Normal	Low	55,526.04	41,303.
rance	Velo	Channel Partners	Normal	Low	12,406.80	9,241.
rance	Velo	Channel Partners	Normal	Medium	21,479.64	15,578.

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What method did you use to replace missing values in numeric fields?

- **Tableau Approach:** Missing values in numerical measures (such as Sales and Profit) were identified using data grid checks and handled using Tableau's native calculated fields. By implementing conditional logic with the formula IFNULL([Sales], 0), any null or unrecorded numeric entry was systematically imputed with a default numeric value (0) to prevent errors during aggregation (e.g., SUM or AVG).
- **Power BI / Power Query Approach:** Using Power Query Editor, the **Column Quality** tool was enabled to visually scan for null or empty percentages across all numeric metrics. Missing numerical fields were then resolved using the **Replace Values** transformation to overwrite null instances with a baseline value or by writing DAX conditional columns (IF(ISBLANK([Sales]), 0, [Sales])).

2. How were outliers handled in your dataset and what visual confirmed their correction?

  - D**etection & Handling Strategy:** Outliers in Sales and Profit were evaluated by defining upper and lower statistical thresholds. Rather than permanently deleting extreme data points—which could skew actual revenue reporting—outliers were handled by creating conditional status flags and capping values. For instance, calculated fields categorized rows as "High Outlier", "Normal", or "Loss" based on specific revenue bounds (e.g., sales exceeding $5,000 in Tableau or $500,000 in Power BI).
    Visual Confirmation:
  - **Box Plots:** In Tableau, a disaggregated Box Plot displaying individual transactions clearly showed the upper whisker boundary, highlighting extreme values as colored dots above the normal distribution fence.
  - **Scatter Plots:** In Power BI, a **Profit vs. Sales Outlier Analysis** scatter plot visually segmented transaction clusters by color, confirming that high-value outliers and negative-profit sales were properly labeled and isolated.
  - **Validation Bar Charts:** A bar chart displaying record counts per status (e.g., 20 High Outliers vs. 10,174 Normal records in Tableau, or 97 High Outliers in Power BI) provided quantitative proof that extreme values were safely isolated.

3. What inconsistencies were detected in categorical values and how were they standardized?

  - **Types of Inconsistencies Detected:** Categorical fields (such as Region, Country, Segment, and Discount Band) exhibited issues including inconsistent text casing, redundant entries, leading/trailing whitespace, and missing values.
- **Standardization Process:**
- **Power Query Text Transformations:** Text fields were standardized using **Trim** (to eliminate invisible spacing errors) and **Clean** (to remove non-printable characters). The **Replace Values** feature was applied to map null text cells to "Unknown".
- **Grouping and Aliases in Tableau:** Mismatched naming variations (e.g., unifying regional naming conventions) were resolved using Tableau's **Edit Aliases** and **Create Group** functions to merge duplicate categories into standardized, uniform labels prior to visual analysis.

4. Describe how calculated fields helped in resolving data issues.

Dynamic Data Imputation: Calculated fields enabled on-the-fly handling of missing data (IFNULL) without permanently altering the underlying raw data files, ensuring the original dataset remains untouched while serving clean data to the visuals.
Rule-Based Categorization & Segmentation: They allowed the creation of custom business logic to convert raw numerical measures into structured categorical dimensions (such as Profit_Status or Sales Outlier Flag).
Outlier Capping & Thresholding: Through conditional logic (IF...THEN...ELSE), calculated fields allowed us to cap extreme values at an upper boundary for specific analyses, ensuring that extreme data points do not distort visual scales or skew baseline averages.

5. How did your dashboard demonstrate the effectiveness of the data cleaning process?

- **High-Level Executive Metrics (KPIs):** Summary KPI cards prominently displayed metrics such as **100% Data Completeness** and total field/record counts, proving that all missing entries were successfully resolved.
- **Before vs. After Validation Visuals:** Side-by-side comparative bar charts (comparing raw Sales vs. Cleaned Sales) visually verified that data transformation logic retained total volume and overall distribution without introducing unintended data loss.
- **Transparent Outlier Profiling:** Dedicated visual breakdowns—such as distribution histograms, profit scatter plots, and outlier proportion donut charts (e.g., showing 77.86% Normal, 13.86% High Outlier, and 8.29% Loss)—demonstrated that data anomalies were thoroughly audited, flagged, and accounted for prior to drawing business insight

RESULT:

The dataset was successfully cleaned and prepared for accurate analysis using Tableau and Power BI. Issues such as nulls, inconsistencies, and outliers were identified and resolved, improving data quality significantly.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		